Agenda

- Intro to ML + Notation
- Bias, Variance
- Curse of dimensionality

1. Intro

1.1

Y : random response/target

$X : (X_1, X_2, \dots, X_p)$ random predictor vector

$\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$

$\mathbb{X}$ = $n \times p$ data matrix

p = # of predictors (Not βs)

n = # of obs (training set)

x_0 : test point

x : a fixed data point

x_i : i^{th} training point

$\beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$ β_0 = intercept

$\mathcal{T}$: training set $\{(x_1, y_1), \dots, (x_n, y_n)\}$

Assumption $Y = f(X) + \varepsilon$ want to estimate f

1.2 Prediction: $\hat{Y} = \hat{f}(X)$ want $\hat{f}$ to be close to Y

this class

Inference: how does Y depend on X? $\hat{f}$ is the answer.

1.3 Error

ε not predictable/observable $\quad Var(\varepsilon) = \sigma^2$

$X = x$ (i.e for some observed point x)

$$E\left[(Y-\hat{Y})^2 \,\middle|\, X=x\right] = \left[f(x) - \hat{f}(x)\right]^2 + \sigma^2$$

make this small

σ^2: can't measure (irreducible)

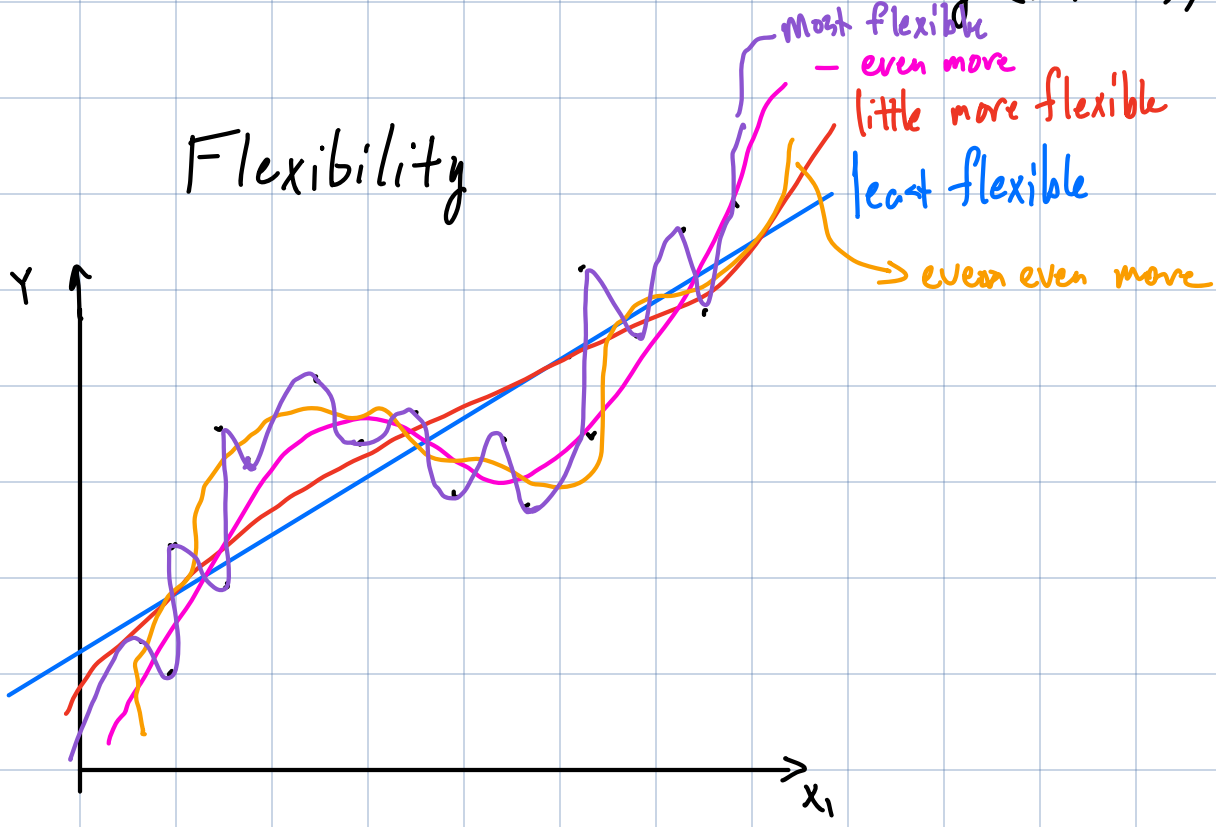
1.4 Estimating f

- Parametric: e.g. $f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$
  Goal: estimate βs
- Non-parametric: no functional form assumption

1.5 Types of learning

Types	Response/Target	Algorithms/Model
Supervised		
↳ Regression	Y is quantitative	ridge, lasso, splines, trees
↳ Classification	Y is qualitative	logistic, trees, naive Bayes
Unsupervised	None	Clustering (k-means), PCA

1.6 Flexibility



Most flexible ≠ best model

2 What is f?

2.1 Loss $L(Y, f(X))$

Regression: Square error $(Y - f(X))^2$

We want to minimize expected prediction error

$$EPE(f) = E\left[(Y - f(X))^2\right]$$

minimized by $f(x) = E[Y \mid X = x]$ — Conditional mean of Y given $X = x$

2.2 Our goal is approximate $E[Y \mid X = x]$

(A) Linear regression

(B)

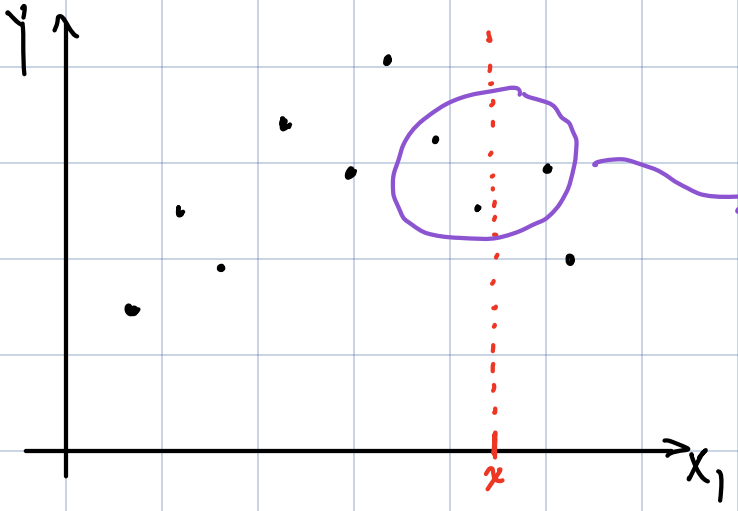
ft^2	rent
900	\$2k
1000	\$2.2k
1700	\$3k
1800	\$3.1k
2500	\$5k

How much rent for $1400\ ft^2$?

K-Nearest Neighbors (kNN)

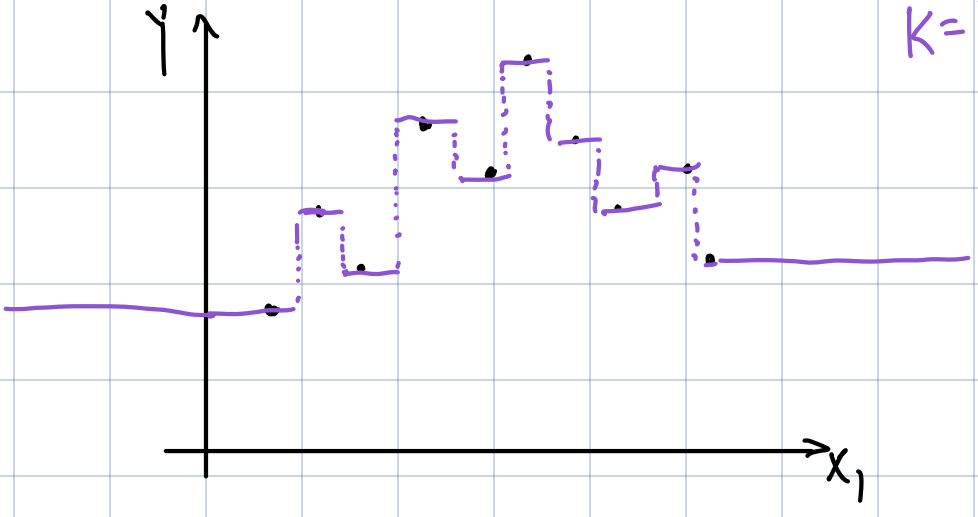
$$\hat{f}_K(x) = \frac{1}{K} \sum_{i : x_i \in N_K(x)} y_i$$

$N_k(x)$ = k training points closest to x



K=3

take $\bar{y}$ of these 3 pts as our $\hat{f}(x)$



K=1

3. Training vs Testing Error

Squared error

Training error $\quad \overline{err} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$

Test error of your fitted model $\quad Err_{\mathcal{T}} = E\left[\left(Y_0 - \hat{f}(X_0)\right)^2 \middle| \mathcal{T}\right]$

(X_0, Y_0) new random data point

4. Bias-variance decomposition

Fix x_0. $\hat{f}(x_0)$ is random since $\mathcal{T}$ is random.

$$\bar{f}(x_0) = E\left[\hat{f}(x_0)\right]$$

avg the prediction over all training sets

Theorem 4.1

$$E\left[(Y_0 - \hat{f}(x_0))^2\right] = \sigma^2 + \left[f(x_0) - \bar{f}(x_0)\right]^2 + E\left[(\hat{f}(x_0) - \bar{f}(x_0))^2\right]$$

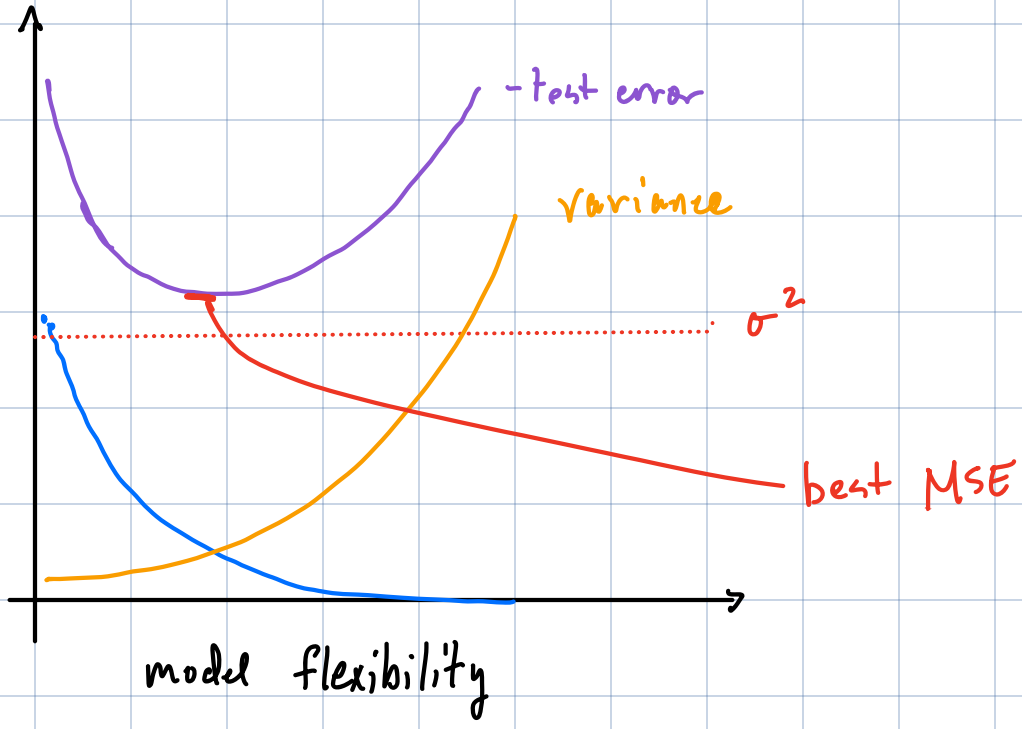
σ^2 is irreducible error

$\left[f(x_0) - \bar{f}(x_0)\right]^2$ is bias^2 of $\hat{f}(x_0)$

how far avg fit from truth

$E\left[(\hat{f}(x_0) - \bar{f}(x_0))^2\right]$ is variance of $\hat{f}(x_0)$

how much $\hat{f}(x_0)$ varies between training sets





low bias
low variance

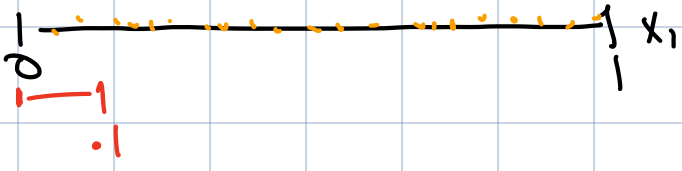
high bias
low variance

low bias
high variance

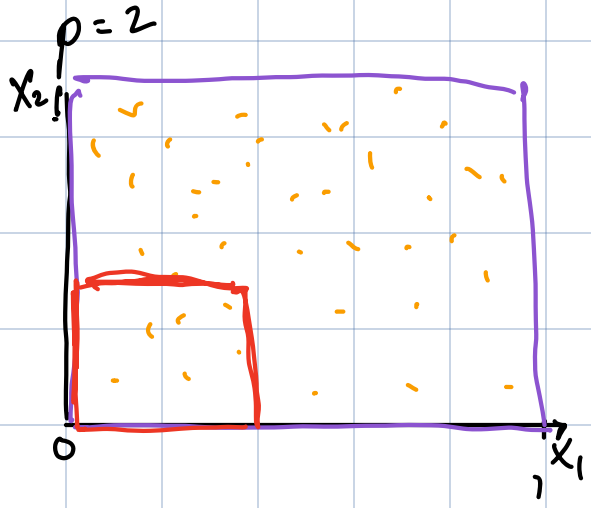
5. Curse of Dimensionality

In high dimensions (p) neighborhoods are not local

$p = 1$



we capture 10% of data



Fraction of data = fraction of volume

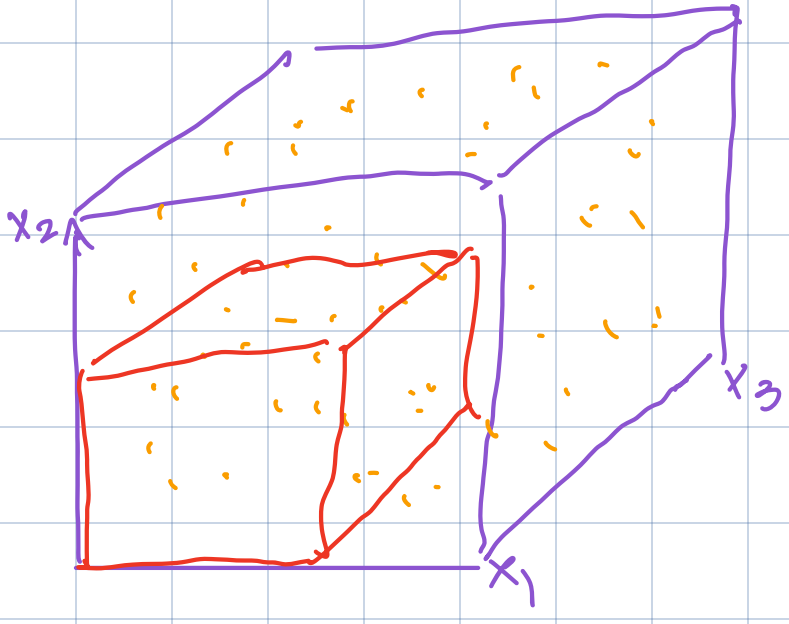
e = edge length

if p = 2, volume = $e^p = e^2$

$e^2 = .10$

$e = \sqrt{.10} \times .32$

p = 3



p = 3

$e = .1^{1/3} = .46$

p	segment length to capture 10%
1	.1
2	.32
3	.46
10	.79
100	.98